

Design Patent Similarity with ArcFace Metric Learning

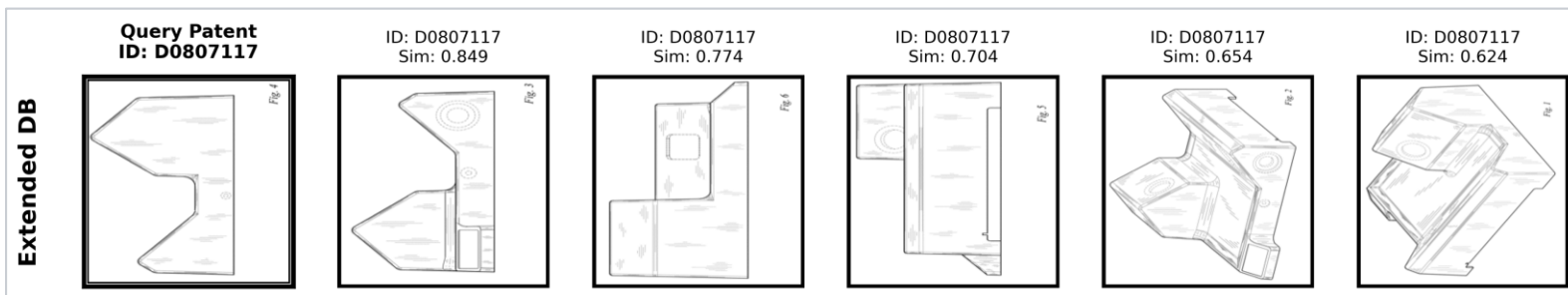
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Patent identity—not product category—is the right supervision for finding visually similar designs.

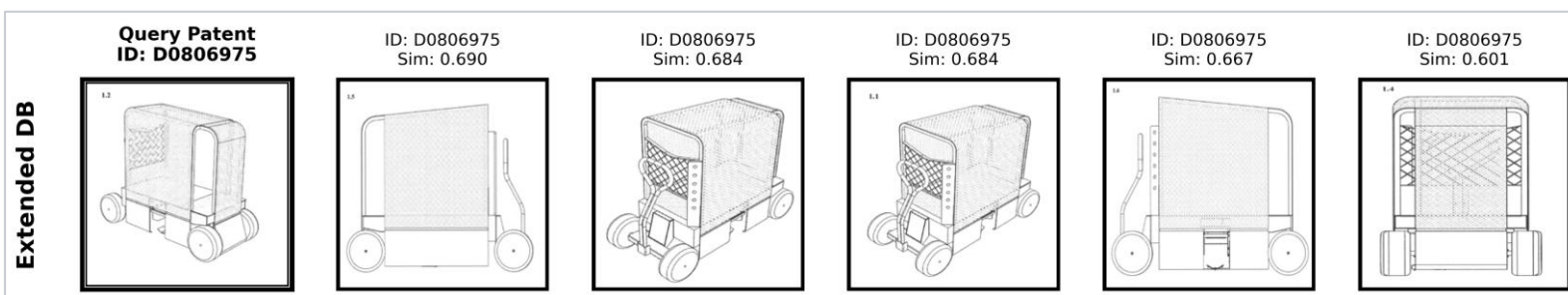
15-MINUTE RESEARCH TALK

Three retrieval examples show success and hard negatives



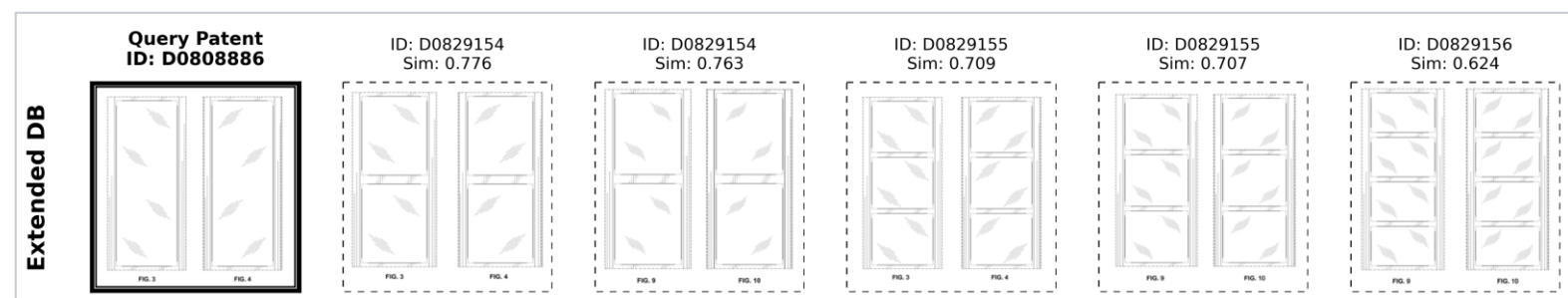
Example 1 · exact recovery

Query D0807117: all five retrieved views share the query patent ID.



Example 2 · exact recovery

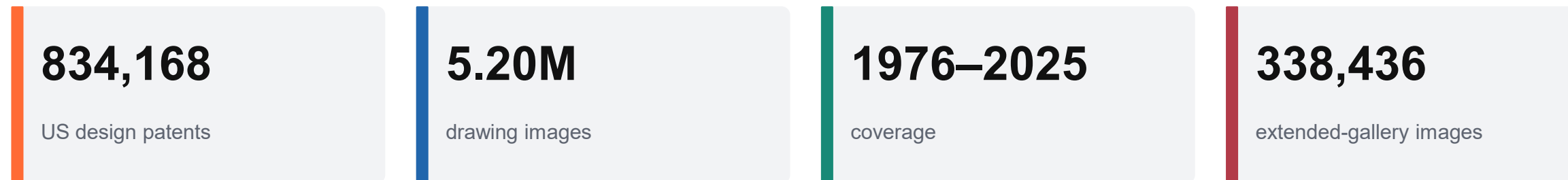
Query D0806975: all five retrieved views share the query patent ID.



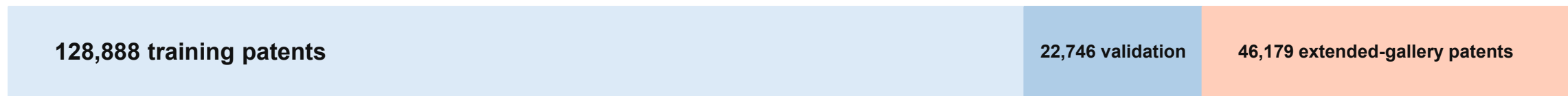
Example 3 · hard negative

Query D0808886: visually close panels rank highly despite different patent IDs.

A national-scale corpus makes the task realistic



Train / validation / retrieval split



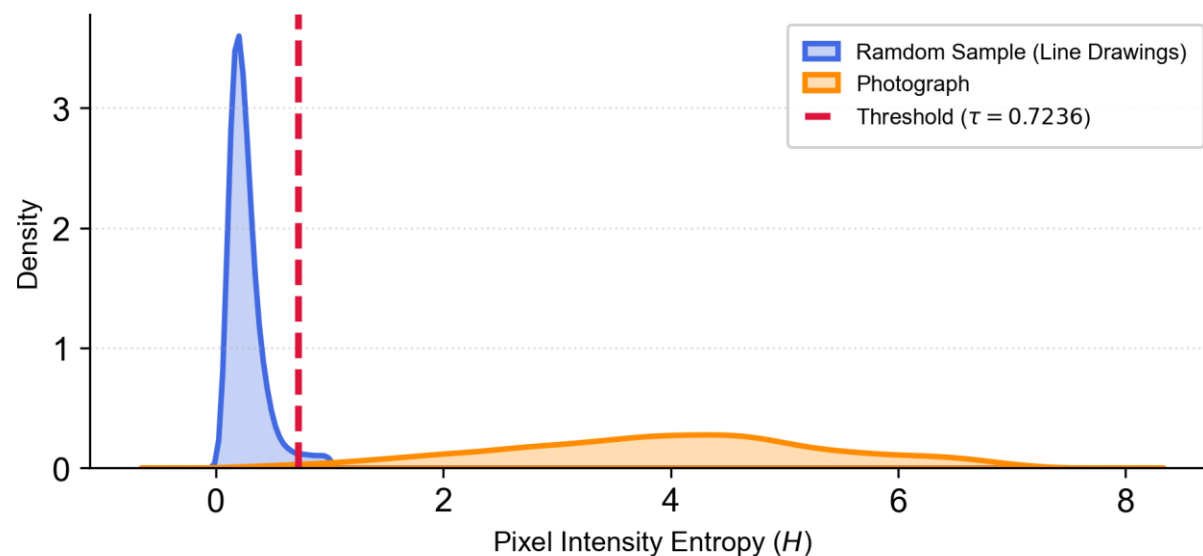
The extended gallery is 8.7× larger than the standard gallery—closer to production retrieval.

PhotoLike filtering removes photographs before training

Representative photographic-style patents



Pixel-intensity entropy separates the domains



Line drawings are sparse and low-entropy;
photographs contain richer shading and texture.

Threshold

$\tau = 0.724$
99% target recall

Flagged

28,717
candidates

Excluded

25,448
verified patents

Patent IDs become the metric-learning signal



Training

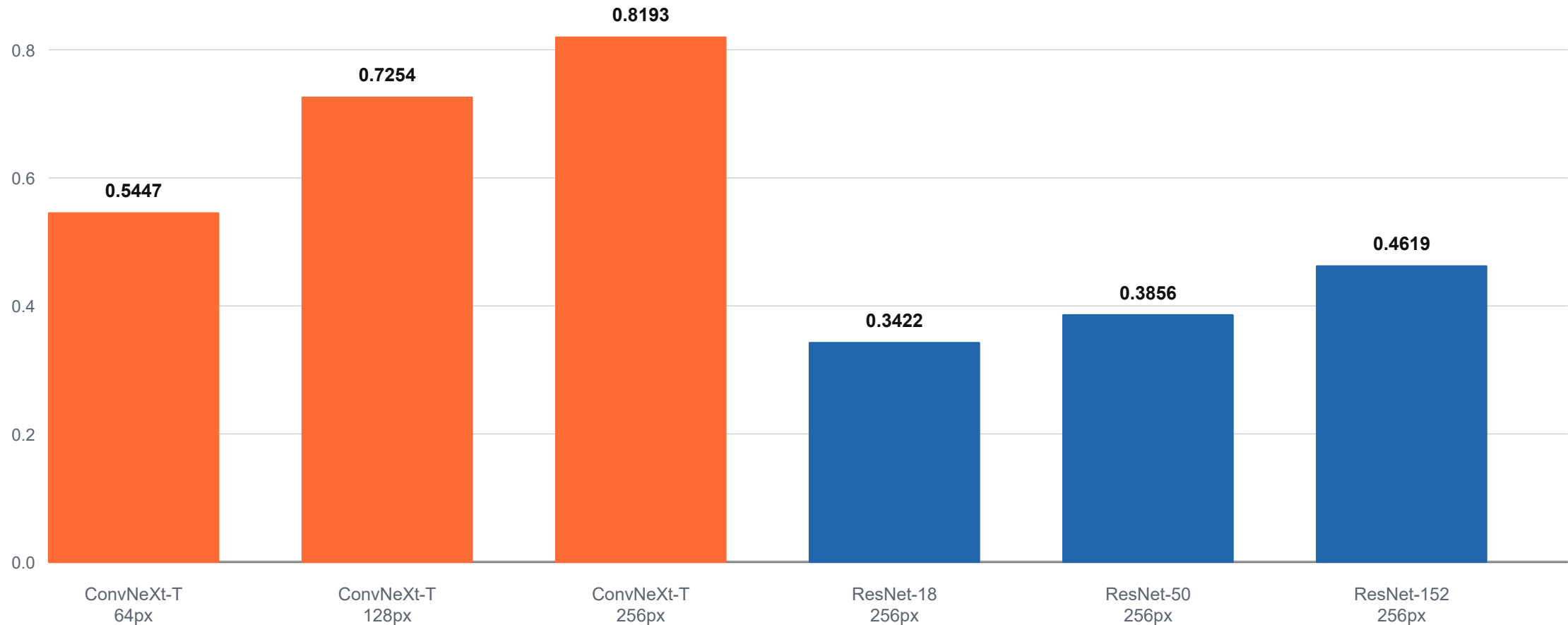
Scale 30 · angular margin 0.50 · 30 epochs · effective batch ≈ 512

Retrieval

Cosine search, then optional query expansion or graph diffusion

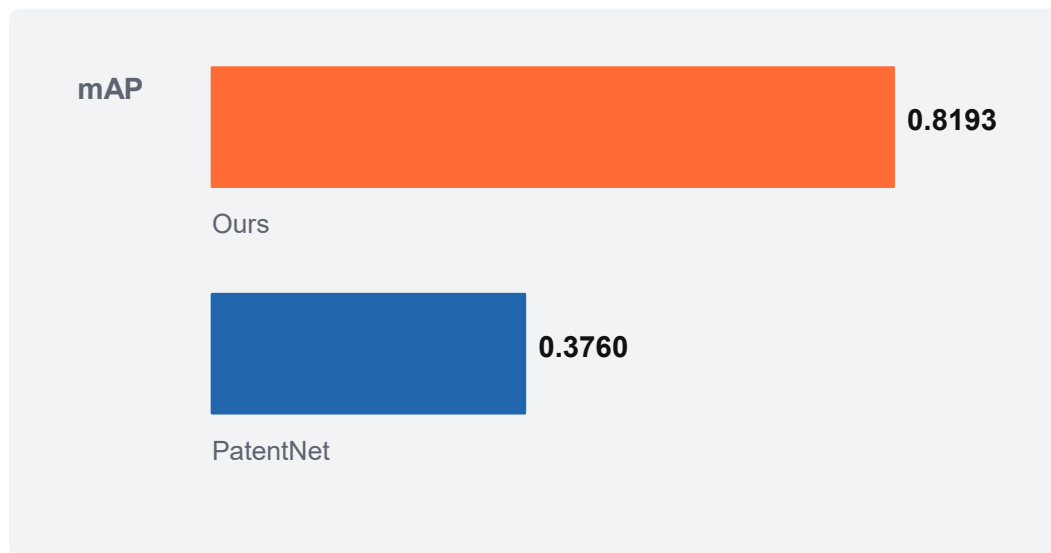
Resolution and backbone capacity both matter

BEST: 0.8193 mAP



The gain survives a much larger search gallery

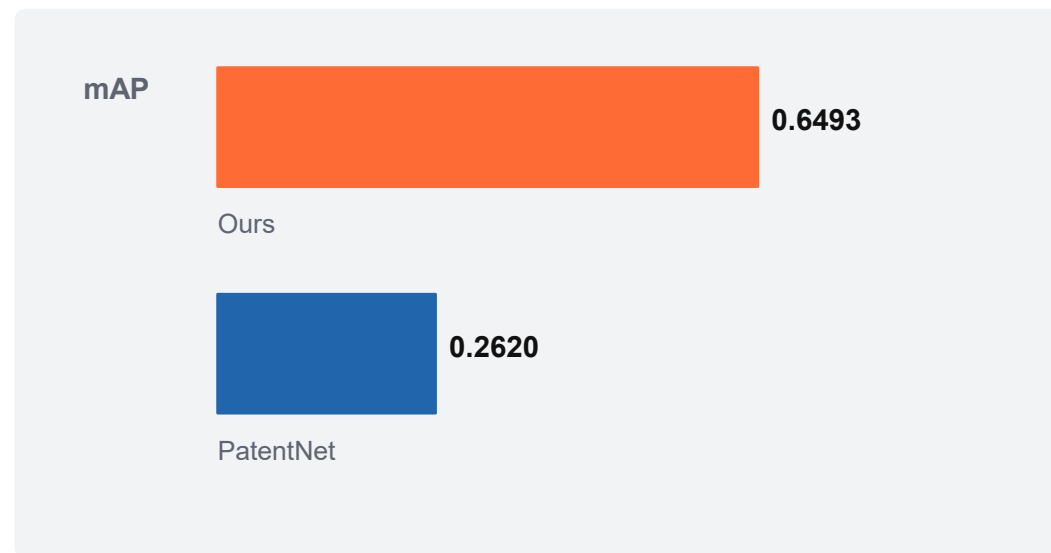
Standard Gallery



Acc@1 = 0.9312

Acc@20 = 0.9906

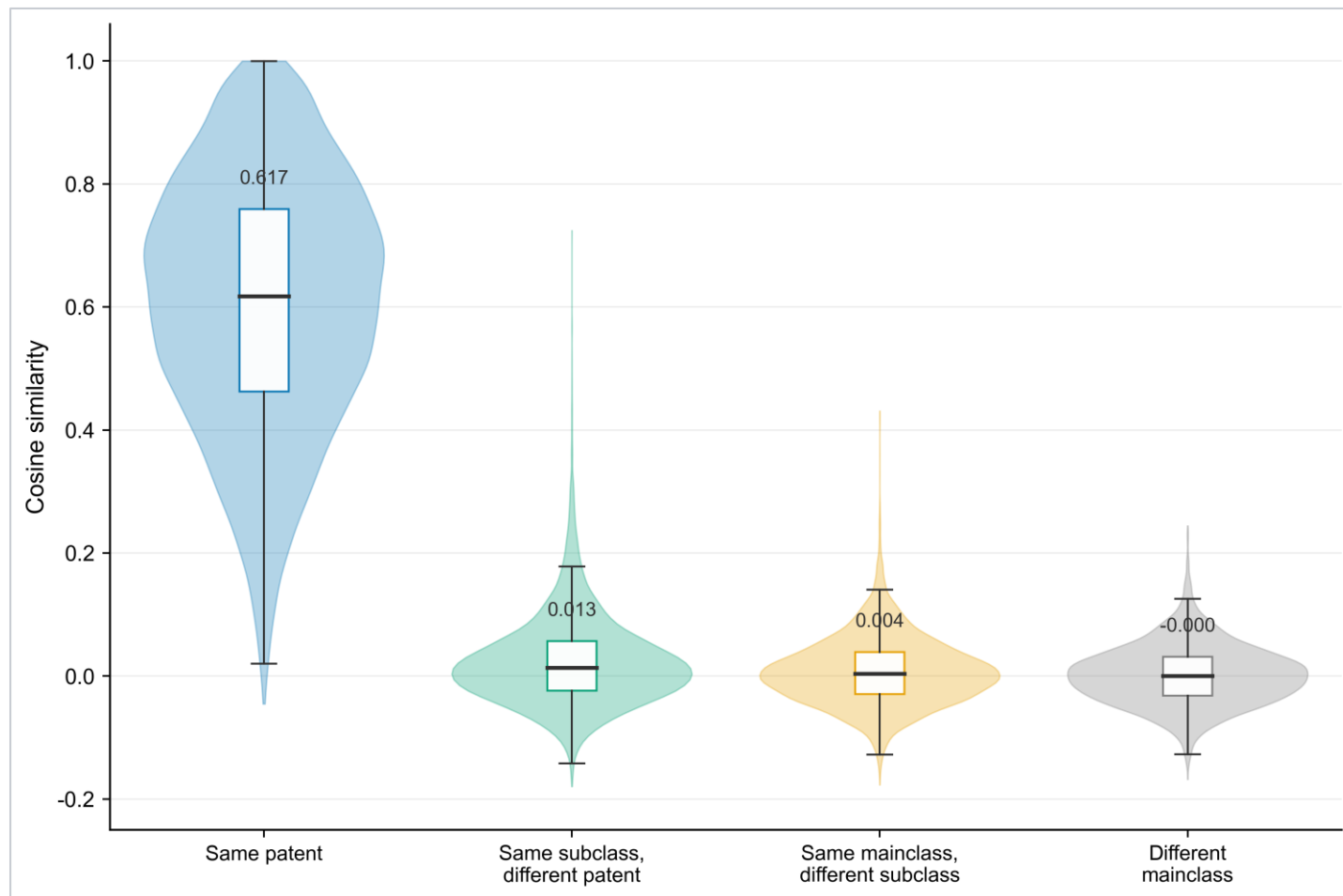
Extended Gallery



Acc@1 = 0.8088

Acc@20 = 0.9711

The embedding learns identity—not just taxonomy



0.617

median: same patent

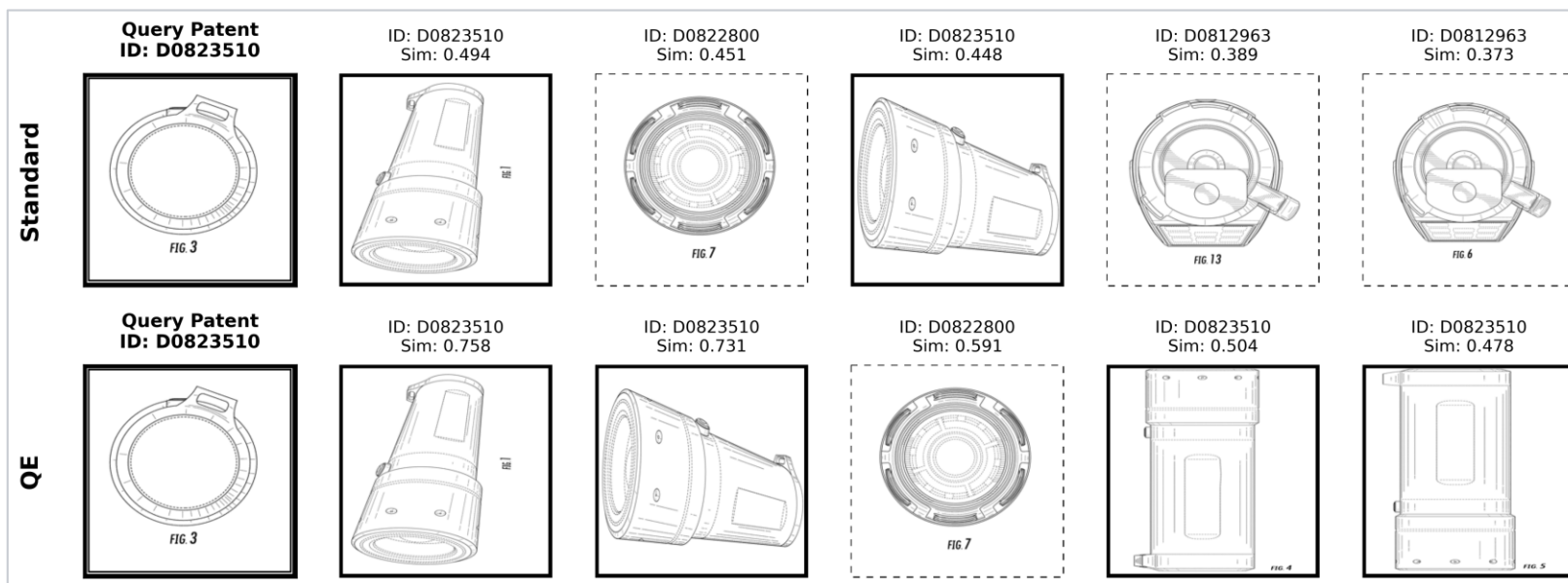
0.013

same subclass, other patent

≈0

different main class

Query expansion improves ranking—selectively



0.8193

baseline mAP

0.8417

query expansion mAP

Best setting

$k = 3 \cdot \alpha = 1$

Diffusion peaks at 0.8332.

A deployable retrieval layer—with clear limits

1 · Supervision

Patent identity turns multi-view drawings into strong positive pairs and avoids category-only shortcuts.

2 · Scale

High-resolution ConvNeXt features maintain useful precision in a 338k-image extended gallery.

3 · Next step

Hard negatives, domain filtering, and examiner-centered evaluation remain essential.

Bottom line: metric learning can narrow prior art; it does not replace legal judgment.